

Perceptual Motor Learning for Zero-Shot Generalization in Autonomous Lateral Control

Elahe Delavari¹, John Moore², Junho Hong¹, and Jaerock Kwon¹

Abstract—Generalization to unseen environments remains the central unsolved challenge for autonomous driving systems. This paper presents a *Perceptual Motor Learning* (PML) framework inspired by the Active Inference principle of prediction-error minimization, enabling robust lateral control without retraining. PML learns a forward generative model that predicts the sensory consequences of steering actions and selects actions whose predicted outcomes best match a preferred visual state. Implemented as a U-Net–based transition model evaluated via the Structural Similarity Index, the system requires only a monocular visual input and no privileged sensing (no LiDAR, GPS, or HD maps). Trained exclusively in one CARLA simulator town, PML achieves 100% success on straight, one-turn, and two-turn tracks in two entirely unseen towns, outperforming Behavioral Cloning and Proximal Policy Optimization-based Reinforcement Learning baselines in cross-domain transfer. On the CoRL2017 benchmark, PML delivers 92% success in an unseen test town despite training only in a structurally disjoint third town. These results indicate that compact predictive world models grounded in prediction-error minimization provide a principled, data-efficient path toward zero-shot generalization in autonomous driving.

Index Terms—Autonomous Driving, Generalization, Perceptual Motor Learning, Active Inference, World Models, CARLA

I. INTRODUCTION

A fundamental barrier to deploying autonomous vehicles (AVs) is the inability of learned policies to generalize beyond their training distribution. Modular Pipelines (MPs) [1] suffer from error propagation across stages; Imitation Learning (IL) [2], [3] faithfully reproduces expert behavior but collapses on out-of-distribution observations; Reinforcement Learning (RL) [4] requires costly reward engineering and extensive environment interaction. All three paradigms implicitly assume generalization comes from *more data* or *finer tuning*—rather than from a richer internal causal model of the world.

This workshop asks a sharp question: *how can AV systems generalize to new environments or unseen conditions?* We argue the answer requires a shift in representational substrate: agents should learn *causal action-observation dynamics* rather than static input-output mappings. Such a representation is, by construction, more transferable because road-following dynamics (“steer left $\Rightarrow$ road shifts rightward in the image”) are largely invariant across geographies, while appearance statistics are not.

*This work was supported in part by the Ford/University of Michigan Research Alliance Program (10.13039/100007270) and in part by the National Science Foundation (NSF) under Grant MRI 2214830.

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Human drivers do not retrain between cities. They carry a learned *predictive model* of how actions cause perceptual changes and transfer that model to novel road structures without explicit supervision. This capability is grounded in *Perceptual Motor Learning* (PML) [5], [6]—the developmental integration of sensory and motor representations into a unified anticipatory loop. We propose that this same principle should underpin autonomous vehicle control.

We present a PML framework for AV lateral control implemented as a deep predictive model that learns action-conditioned visual transitions. Grounded in the prediction-error minimization principle of Active Inference [7], the system selects steering actions whose predicted future observations are most similar to a preferred visual state. Critically, the model is *trained once* in a single simulated town and deployed across diverse, unseen environments without fine-tuning.

The key contributions of this work are:

- A PML controller that operationalizes prediction-error minimization for real-time autonomous lateral control using only monocular visual input, with no LiDAR, GPS, or HD maps.
- Empirical zero-shot cross-town generalization in CARLA, outperforming Behavior Cloning (BC) and Proximal Policy Optimization (PPO)-based RL baselines in all unseen domains.
- Competitive CoRL2017 benchmark [8] performance despite training in a structurally disjoint third town never used in evaluation.
- An honest analysis of the gap between the theoretical Active Inference framework and our practical approximation, with a concrete roadmap for tighter integration.

II. RELATED WORK

A. Generalization in Autonomous Driving

Generalization is the central unresolved challenge for AV learning systems [8]. Conditional Imitation Learning (CIL) [9] improves distributional robustness via high-level command conditioning but still degrades under domain shift. Learning By Cheating (LBC) [10] achieves strong benchmark results by training privileged and image-only agents jointly, but its two-stage complexity limits transfer. Conditional Affordance Learning (CAL) [11] maps visual inputs to intermediate representations; its performance depends tightly on the quality of affordance predictions. A recurring pattern across all these methods is that they mitigate the generalization problem without addressing its root cause: policies encode *appearance statistics*, not causal structure.

B. World Models and Predictive Control

Predictive world models offer a principled substrate for generalizable control [12]. Rather than mapping observations directly to actions, these models learn the causal dynamics of the environment, allowing agents to imagine consequences before acting. Action-based representation learning [13] showed that encoding causal action-observation dependencies substantially improves policy transfer. Prior action-conditioned scene prediction [14], [15] focused on standard scenarios in which future frames closely resemble the input; we extend this to a steering-conditioned formulation that requires large visual changes (e.g., entering a curve) to be accurately predicted.

C. Active Inference for Autonomous Agents

Active Inference (AIF) formalizes perception and action as joint minimization of variational free energy [7], [16]. Prior AIF applications to robotics [17] and hierarchical navigation [18] demonstrate its principled appeal, but high computational overhead and the difficulty of scaling probabilistic inference to high-dimensional visual inputs have limited realistic deployment. Nozari et al. [19] combine AIF with IL using Dynamic Bayesian Networks, but the computational cost is prohibitive for resource-constrained AV platforms. Our framework retains the core AIF principle—prediction-error-minimizing action selection—while replacing full probabilistic inference with a deterministic observation-space approximation that is tractable in real time.

III. METHOD

A. Perceptual Motor Learning as Prediction-Error Minimization

PML formalizes the bidirectional mapping between sensory observations and motor actions as a closed-loop process that minimizes perceptual discrepancy. Let observations $o_t \in \mathbb{R}^{d_o}$ and actions $a_t \in \mathbb{R}^{d_a}$ evolve according to an unknown transition $o_{t+1} = \mathcal{T}(o_t, a_t) + \epsilon_t$. The agent learns a forward transition model $f_s: (o_t, a_t) \mapsto \hat{o}_{t+1}$ and selects actions that minimize the discrepancy between predicted and preferred future observations:

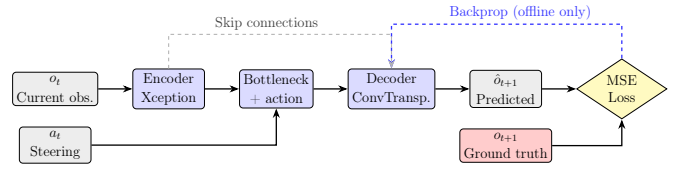
$$a_t^* = \arg \min_{a_t \in \mathcal{A}} f_d(f_s(o_t, a_t), o_{\text{pref}}). \quad (1)$$

The joint learning objective couples forward model training with action consistency:

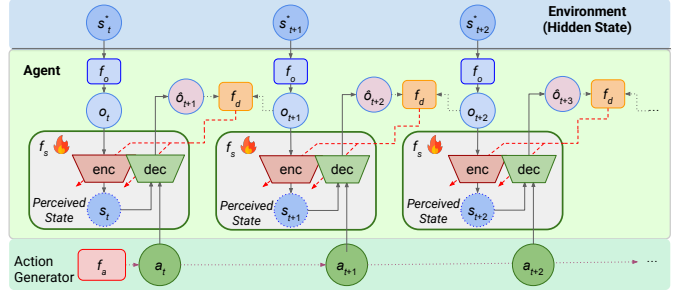
$$\min_{\theta_s, \theta_\pi} \mathbb{E}_{(o_t, a_t, o_{t+1}) \sim \mathcal{D}} [\mathcal{D}(o_{t+1}, f_s(o_t, a_t; \theta_s)) + \lambda \mathcal{L}_a(a_t, \pi(o_t; \theta_\pi))], \quad (2)$$

where $\mathcal{D}(\cdot, \cdot)$ measures perceptual reconstruction error, $\mathcal{L}_a$ is motor consistency loss, and λ balances the two terms. The environment is modeled as a Partially Observable Markov Decision Process (POMDP): the agent observes only visual inputs o_t while the true state s_t remains hidden, requiring inference of hidden state dynamics from sensory data alone.

Connection to Active Inference. This formulation approximates Expected Free Energy (EFE) minimization [7]. Full



(a) Schematic block diagram of the offline training pipeline.



(b) Detailed view of the task-agnostic offline training loop. The hidden environment state s_t^* is perceived via $f_o(\cdot)$; the forward model $f_s(\cdot)$ (enc+dec with skip connections, shown by the flame icon during training) learns to predict $\hat{o}_{t+1}$ from (o_t, a_t) ; the distance model $f_d(\cdot)$ supplies the MSE training signal via the red dashed backpropagation path.

Fig. 1: Offline training of $f_s(\cdot)$. (a) Block-level overview; (b) unrolled temporal diagram showing the full perception–action loop across three consecutive timesteps. The model is *frozen* after this phase and never updated during deployment.

AIF maintains a posterior over latent states and selects policies minimizing the KL divergence between predicted and preferred state distributions:

$$G(a_t) = \mathbb{E}_{q(o_{t+1}, s_{t+1}|a_t)} [D_{\text{KL}}(q(s_{t+1}|a_t) \| p_{\text{pref}}(s_{t+1})) - \mathcal{H}(q(o_{t+1}|s_{t+1}, a_t))]. \quad (3)$$

where $D_{\text{KL}}(q(s_{t+1}|a_t) \| p_{\text{pref}}(s_{t+1}))$ encourages the selection of actions that lead to preferred or desired states, and $\mathcal{H}(q(o_{t+1}|s_{t+1}, a_t))$ represents the entropy of predicted observations, corresponding to uncertainty reduction that guides exploration. Our approach replaces the latent-state posterior with direct observation-space predictions and substitutes the KL divergence with SSIM-based image similarity, approximating the *pragmatic* (goal-directed) term of EFE while discarding the *epistemic* (information-gain) term, which provides little benefit once the forward model is fixed offline. We emphasize this is a *principled simplification*, not a full AIF implementation; Section V returns to this distinction.

B. Forward Transition Model

The forward model $f_s(\cdot)$ is a U-Net with an Xception-style encoder [20], [21]. Skip connections preserve spatial road structure across encoding and decoding paths—critical for pixel-level prediction—and the encoder-decoder structure implicitly learns latent representations that support prediction-error-based decision making.

Given $o_t \in \mathbb{R}^{160 \times 160 \times 1}$ (road-only grayscale) and candidate steering a_t , the network predicts $\hat{o}_{t+1}$. The action is injected

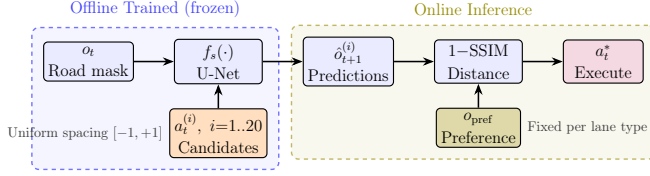


Fig. 2: PML control loop. At each timestep, 20 candidate steering actions are evaluated through the frozen $f_s(\cdot)$. The action whose predicted future observation $\hat{o}_{t+1}$ is most similar (SSIM) to o_{pref} is selected and executed.

into the bottleneck via Dense layers (ELU activation) before decoding; `conv2d_transpose` upsampling reduces blurriness relative to bilinear upsampling. Training uses MSE loss over 72,000 augmented image-action-next-image triplets from **Town06**, collected via a zigzag exploration policy that ensures uniform coverage of steering values $[-1.0, +1.0]$. Horizontal flipping with negated labels doubles the dataset at no additional collection cost. The model is *completely frozen* after training and requires no gradient updates during deployment.

C. Action Selection (Covert Simulation)

At each control timestep, $d_A = 20$ steering values uniformly spaced on $[-1.0, +1.0]$ are each fed to the frozen $f_s(\cdot)$. The fourth predicted future frame is used (providing a longer look-ahead that better captures the visual consequence of turns). Each prediction is scored against the task preference:

$$a_t^* = \arg \min_{i \in [0, \dots, d_A-1]} [1 - \text{SSIM}(\hat{o}_{t+1}^{(i)}, o_{\text{pref}})]. \quad (4)$$

SSIM evaluates luminance, contrast, and structure simultaneously, making it more sensitive to spatial road geometry than pixel-wise MSE. Speed is fixed (throttle = 0.2) to isolate lateral control. Fig. 2 shows the complete PML control loop.

Preference design. The preference image o_{pref} is a road-only grayscale segmentation of a straight, well-centered lane. Lane-specific preferences are used when the road structure differs significantly between towns (e.g. Town01 with two lanes, as Town04 and Town06 are multi-lane). Images used as preferences for the PML agent are presented in Fig. 3. No preference is adapted online; all are fixed before deployment. Fig. 4 illustrates the full two-timestep unrolled action-selection loop, showing how the frozen $f_s(\cdot)$ evaluates all candidate actions in parallel and how the MUX selects the preferred action at each step.



Fig. 3: The examples of the preferences. The first is for Town01 and Town02. The center two are for lanes 1 and 2 in Town04. The right two are lanes 3 and 4 in Town06.

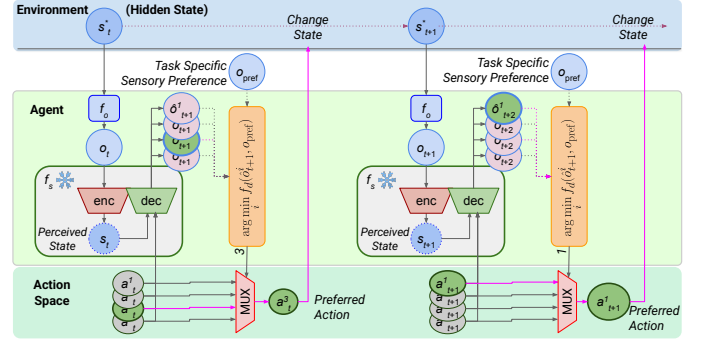


Fig. 4: Detailed online action-selection loop unrolled over two timesteps. At each step, the frozen $f_s(\cdot)$ (snowflake icon) generates one predicted future observation per candidate action. The $\arg \min_i f_d(\hat{o}_{t+1}^{(i)}, o_{\text{pref}})$ block scores every prediction against the task-specific sensory preference; the MUX routes the winning action to the environment, which transitions to a new hidden state.

IV. EXPERIMENTS

A. Setup

All experiments use CARLA [8]. PML is trained exclusively in **Town06** (multi-lane highway with many entrances and exits) and evaluated zero-shot in **Town01** (bridges and a river), **Town04** (mountain roads with a loop highway), and **Town06** itself. Three track types per town are tested: straight [A], one-turn [B], and two-turn [C], each over four independent runs (Fig. 5). **Towns 01** and **04** represent domain shift: neither geometry, road width, nor visual structure matches **Town06**.

Three baselines are evaluated, all trained in **Town06**: **BC**—a five-layer CNN trained via behavioral cloning on 47,313 normalized steering samples; **RL-RGB**—PPO [22] trained for 4M steps with RGB input; **RL-Gray**—PPO trained for 4M steps on semantic road masks. All share a 20-bin discrete action space and fixed throttle of 0.2. No agent is fine-tuned before evaluation in new towns.

The RL reward function encourages stable lane-keeping, smooth steering, and consistent heading alignment:

$$r_t = 0.2 - 1.0 \min(d_{\text{lane}}, 2.0) - 0.3 \min(|\psi|, 0.3) - 0.02 |\Delta \text{steer}|, \quad (5)$$

where d_{lane} is lateral deviation (m), ψ is heading error (rad), and Δsteer is the change in steering between steps. Episodes terminate with a -5.0 penalty if $d_{\text{lane}} > 2$ m.

Metrics: **Average lateral deviation**—mean Euclidean distance (m) from predefined waypoints—and **success rate**—proportion of runs completed within 0.5 m of lane center over 600 simulation steps.

B. Implementation Details

PML. Adam optimizer, $1r10^{-4}$, batch 128, 50 epochs, NVIDIA RTX 6000 (48 GB). Inference: 20 forward passes per control step at ≈ 10 Hz on an RTX 4060 (8 GB).



Fig. 5: Test tracks in Town01 (left), Town04 (center), and Town06 (right). Yellow arrow [A] = straight; green arrow [B] = one-turn; cyan arrow [C] = two-turn. PML is trained in Town06 only; Town01 and Town04 are entirely unseen at test time.

Algorithm 1 PML Lateral Control (Online)

Require: Frozen model f_s , preference o_{pref} , action set $\mathcal{A} = \{a^{(i)}\}_{i=1}^{20}$, look-ahead $k = 4$

- 1: **while** driving **do**
- 2: Capture current observation o_t (road-only mask)
- 3: **for** each candidate $a^{(i)} \in \mathcal{A}$ **do**
- 4: $\hat{o}^{(i)} \leftarrow f_s^k(o_t, a^{(i)})$ // k -step rollout
- 5: $d^{(i)} \leftarrow 1 - \text{SSIM}(\hat{o}^{(i)}, o_{\text{pref}})$
- 6: **end for**
- 7: $a_t^* \leftarrow \mathcal{A}[\arg \min_i d^{(i)}]$
- 8: Execute a_t^* ; receive o_{t+1}
- 9: $o_t \leftarrow o_{t+1}$
- 10: **end while**

BC. Conv2D stack (32–64–128–256, stride 2) with batch normalization and 20% dropout, then Dense (512–128–64–1). Adam, lr 10^{-3} , batch 64. Input: $160 \times 160 \times 3$ RGB.

RL Baselines. Stable-Baselines3 [23], trained for 4M steps. RGB PPO: Same network structure as BC network but removed dropout layers. The 512-dim latent convolutional features used as a shared embedding used for the actor–critic heads ;Gray PPO: compact 3-layer CNN, (32–64–128 filters) with stride-2 and ReLU activations, followed by a single fully connected layer producing a 128-dimensional embedding. Both: $\gamma=0.99$, $\lambda=0.95$, batch 128, $n_{\text{steps}}=2048$.

C. PML Inference Algorithm

Algorithm 1 gives the complete runtime procedure. The key property is that the forward model f_s is called *once per candidate action* per timestep, with all other operations being simple array comparisons. This makes inference time proportional to the number of candidate actions rather than environment complexity, giving a predictable real-time budget regardless of town or road type.

D. Cross-Town Generalization

Table I and Fig. 6 summarize the results. PML achieves 100% success in both unseen towns and the lowest average trajectory deviation in two of three settings.

The gap is most dramatic in **Town01**: BC achieves only 8.3% success (road structure too dissimilar from **Town06**), RL-RGB achieves 0%, and RL-Gray achieves 33%. PML succeeds 100%

TABLE I: Cross-town performance (all agents trained in **Town06** only). Dev. = mean lateral deviation (m); SR = success rate (%). Best result per metric per town in **bold**.

Agent	Town01		Town04		Town06	
	Dev.↓	SR↑	Dev.↓	SR↑	Dev.↓	SR↑
PML	0.48	100	0.86	100	0.64	92
BC	1.87	8	1.09	67	1.48	92
RL-Gray	0.87	33	0.51	58	1.20	50
RL-RGB	1.76	0	0.70	33	0.56	58

TABLE II: Per-track success rate (%) in unseen towns. PML achieves perfect success across all track types; baselines degrade on turns.

Town	Track	PML	BC	RL-Gray	RL-RGB
Town01	Straight (A)	100	0	100	0
	One-Turn (B)	100	0	0	0
	Two-Turn (C)	100	25	0	0
Town04	Straight (A)	100	100	75	50
	One-Turn (B)	100	0	75	25
	Two-Turn (C)	100	100	25	25

across all track types. Table II reveals the structure of baseline failures: BC and both RL agents achieve 0% on one-turn and two-turn tracks in **Town01**, while PML is perfect across all combinations.

The PML forward model captures the *causal geometry* of steering-induced visual change, which transfers irrespective of scene appearance. RL-Gray’s partial success in **Town01**—better than RL-RGB—further supports this: stripping appearance noise (retaining only road geometry) improves transfer, consistent with PML’s road-only input strategy.

E. CoRL2017 Benchmark

Table III compares PML against published CoRL2017 results. Standard baselines are trained on **Town01** and tested on **Town02**. PML was trained only in **Town06**—structurally distinct from both evaluation towns—and applied without adaptation. Despite this harder evaluation condition, PML scores 96%/92% in **Town01/Town02**, matching MP and CAL and outperforming RL by 18 percentage points. CIRL and LBC achieve higher scores but require two-stage training pipelines that substantially increase complexity and tuning cost. PML’s

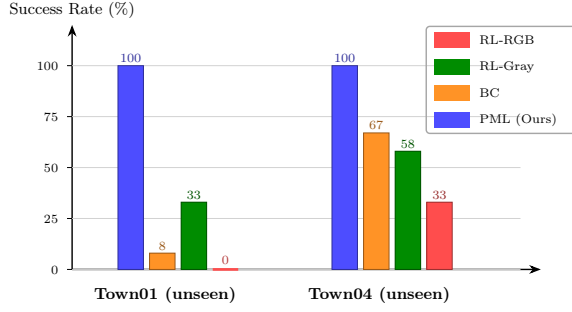


Fig. 6: Success rates (%) in the two unseen test towns. PML achieves a perfect 100% in both; all baselines fall well short, especially in **Town01**.

TABLE III: CoRL2017 straight-driving benchmark, success rate (%). Standard baselines are trained on **Town01**; PML is trained only in **Town06** (never seen **Town01** or **Town02**).

Method	Town01	Town02
MP [8]	98	92
IL [8]	95	97
RL [8]	89	74
CAL [11]	100	93
CIRL [24]	98	100
LBC [10]	100	100
PML (Ours)*	96	92

*Trained in Town06 only; evaluated zero-shot.

TABLE IV: Training cost and complexity comparison.

Method	Training Data	Env. Steps	Retraining?	Stages
BC	47K imgs	—	Required	1
RL-RGB	—	4M	Required	1
RL-Gray	—	4M	Required	1
LBC	large	—	Required	2
CIRL	demos+env	—	Required	2
PML	72K imgs	—	None	1

results come from a single offline training phase on 72,000 images.

As Table IV shows, PML is the only evaluated method that requires neither environment-specific retraining nor a multi-stage training procedure. It trains on fewer images than BC, requires no reward engineering, and generalizes strictly better across domains.

V. DISCUSSION

A. Why Predictive World Models Generalize

PML learns *causal action-observation dynamics*, not a mapping from scene appearance to steering angle. When tested in a new town, the forward model needs only to predict whether a candidate steering action will keep the road centered in view. That causal relationship—“steer left $\Rightarrow$ road shifts rightward in the image”—is invariant across environments; pixel statistics are not. This is the core reason PML transfers where BC fails. BC encodes an implicit mapping from training-town

appearance to steering angle; when **Town01**’s road geometry and visual context differ sharply from **Town06**, the policy degrades catastrophically.

RL-Gray’s partial success in **Town01** (33%) further reinforces this argument. By replacing RGB with road-only semantic masks, RL-Gray removes the appearance noise that causes RL-RGB (0% in **Town01**) to fail under domain shift. PML goes further: it operates in the same semantic road-mask space and additionally couples the perception to a predictive causal model. The covert simulation loop—evaluating all 20 candidate actions at each step—provides a form of online model-predictive control that compensates for local road curvature without any parameter updates. This is the mechanism underlying PML’s 100% success on multi-turn tracks where all baselines fail.

B. Honest Assessment: Limitations and the AIF Gap

We deliberately characterize the gap between this work and a full Active Inference implementation. Three key simplifications were made.

(1) No probabilistic inference. The system maintains no posterior over latent states. A full AIF agent would maintain and update $q(s_t)$ via Bayesian inference at each step, providing calibrated uncertainty that can drive epistemic exploration. Our deterministic model provides none of this.

(2) SSIM as a KL-divergence proxy. SSIM is a perceptually motivated heuristic, not a principled measure of distributional divergence. The EFE’s epistemic term (information gain, $\mathcal{H}$ in Eq. 3) is discarded entirely. A variational formulation using a VAE-based forward model would recover an explicit KL divergence over latent states and restore epistemic foraging—the property that drives exploration toward uncertain regions.

(3) Hand-designed preferences. The preference image o_{pref} is selected manually per lane type. In principle, preferences should be derivable from a higher-level goal specification (a route, a destination, a natural-language instruction) without human engineering. This is an important design debt.

These simplifications make the system tractable and real-time capable on standard GPU hardware. We view them as a concrete research roadmap rather than a fundamental flaw: they define exactly what needs to be added to move from a practical approximation to a principled AIF agent.

C. Relation to the Workshop Themes

PML is neither a data-hungry end-to-end model nor a classical modular pipeline. It represents a third option: a *compact generative model* encoding causal structure that transfers zero-shot. It requires only 72,000 training images—far less than the 4M environment steps needed by RL baselines—and never requires retraining. This positions it directly within the workshop’s exploration of alternatives to scaling as the primary path to generalization.

The approach is also directly relevant to sim-to-real transfer. If the world model captures genuinely causal dynamics (road geometry changes induced by steering), those dynamics should transfer across both domain shift within simulation and the

simulation-to-reality gap. Validating this hypothesis on real-world driving datasets is a central next step.

D. Future Directions

(1) **Longitudinal control**: incorporating throttle and brake, likely requiring a more expressive multi-output forward model; (2) **Dynamic goals**: replacing the static preference image with a goal-conditioned or vision-language-model-derived preference distribution, enabling context-aware driving behavior; (3) **Variational AIF**: implementing a VAE-based forward model to recover true VFE minimization and principled epistemic exploration; (4) **Real-world validation**: testing on nuScenes, Waymo, or collected data to assess the sim-to-real transfer of the causal model; (5) **Continuous action**: differentiable action selection to replace discrete enumeration, enabling smoother and more human-like steering.

VI. CONCLUSION

We presented Perceptual Motor Learning (PML), a prediction-error-minimizing framework for autonomous lateral control that generalizes zero-shot to unseen environments without retraining. Trained on 72,000 images in one CARLA town, PML achieves 100% success on all track types in two structurally distinct unseen towns, outperforming both BC and PPO-based RL baselines. On the CoRL2017 benchmark, PML matches much more complex baselines despite the harder condition of training in a disjoint third town.

We were deliberate in characterizing what PML is and is not: a principled approximation of Active Inference that captures prediction-error-minimizing action selection while foregoing full probabilistic inference. This gap defines a clear research agenda whose pursuit would bring the theoretical guarantees of AIF to bear on the practical problem of AV generalization.

The results suggest that *how* an agent represents its world matters more than *how much* data it sees. A compact causal model of action-conditioned visual dynamics, learned from fewer than 100,000 examples, generalizes more reliably than either imitation of expert trajectories or extensive trial-and-error in a fixed environment. This points toward data-efficient, principled world modeling as a core component of any AV system designed for the open world.

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